

Business Insights Document

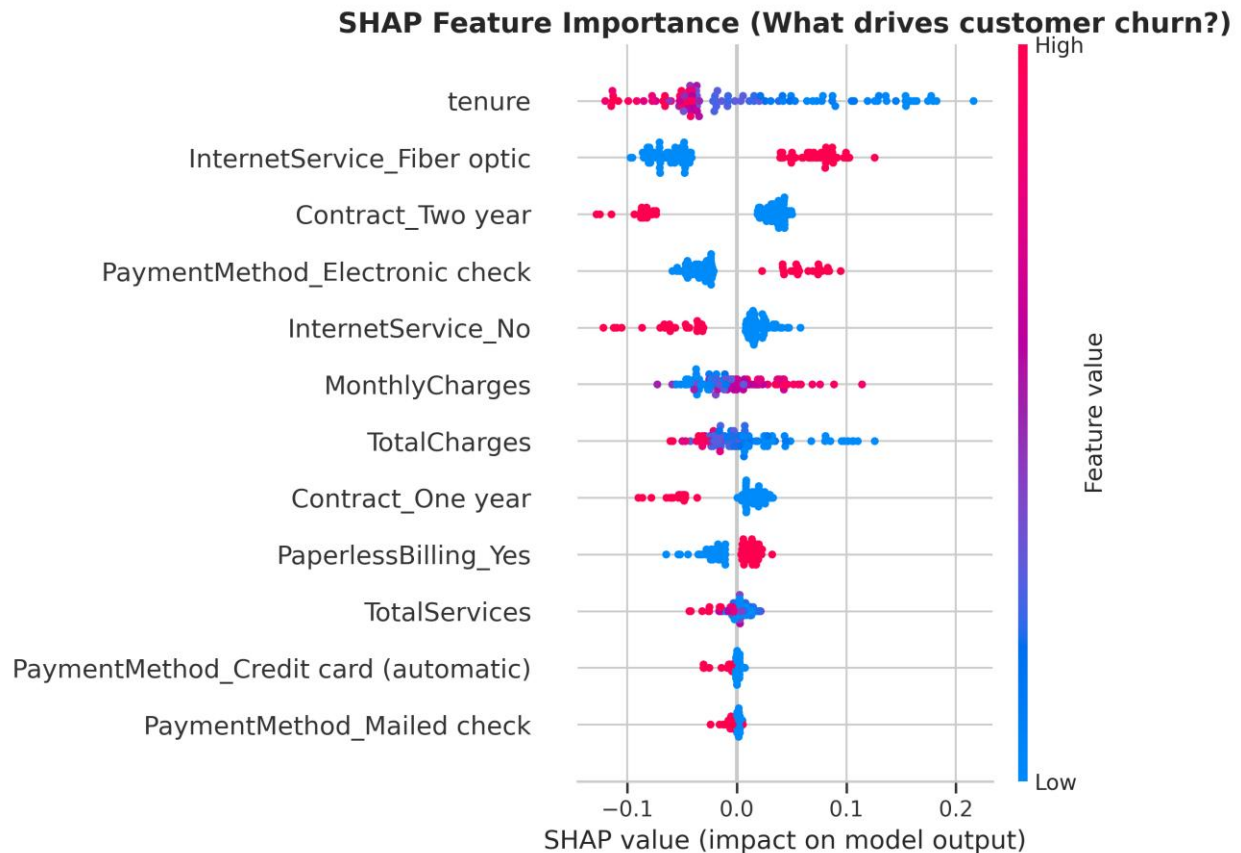
1. Executive Summary

This document outlines the key business insights derived from the customer churn prediction pipeline. By combining Exploratory Data Analysis (EDA) with advanced Machine Learning Explainability (SHAP), we have transformed raw behavioral data into actionable strategies that protect Monthly Recurring Revenue (MRR).

2. Top Churn Drivers (SHAP Explainability Insights)

To understand exactly *why* our model flags a customer as a churn risk, we utilized SHAP (SHapley Additive exPlanations) for model interpretability. The analysis revealed the following top drivers:

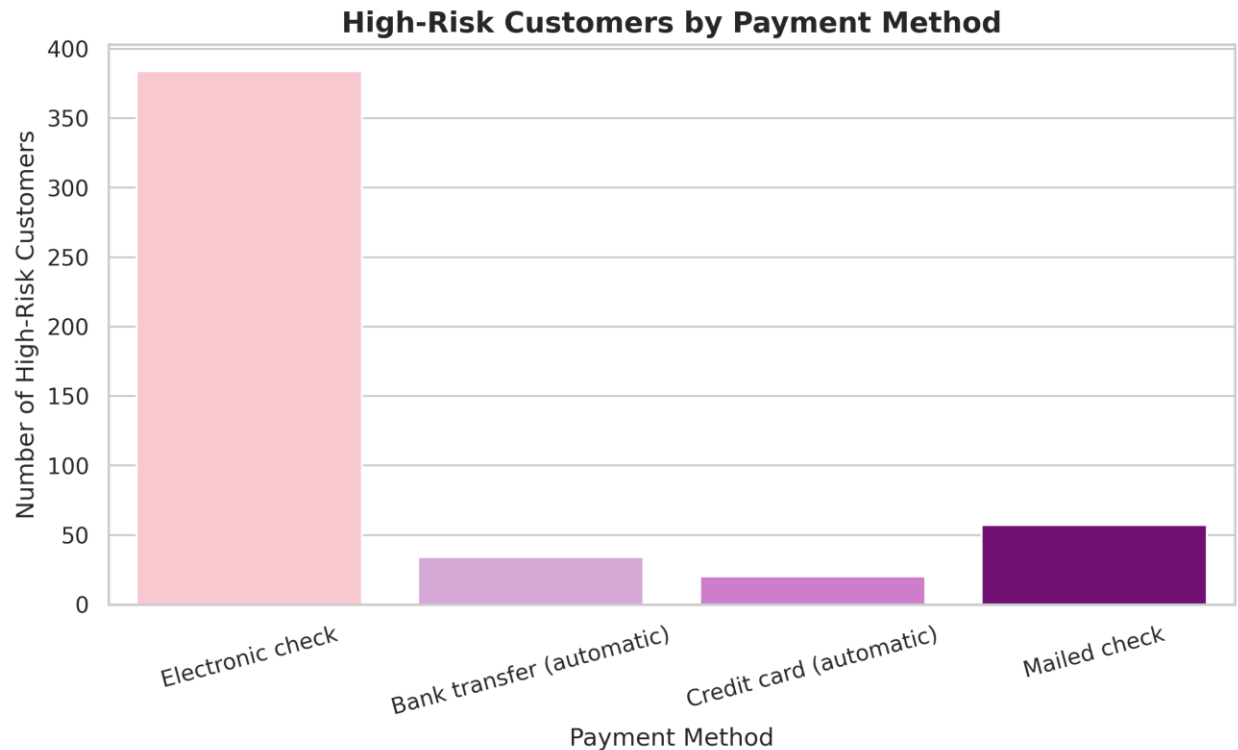
- **Tenure:** This is the most critical feature. The SHAP plot indicates that a shorter tenure strongly pushes the model towards predicting a churn. Customers are most vulnerable in their first few months.
- **Contract Flexibility:** Customers on **Month-to-Month contracts** have the highest churn probability. The lack of long-term commitment removes the barrier to exit.
- **Payment Friction & Charges:** Customers paying manually via **Electronic Checks** and those with high **Total Charges** but fewer supplementary services are highly likely to leave.



3. The "High-Risk" Customer Profile

Based on our predictive risk modeling and EDA, a "High-Risk" customer typically exhibits the following traits:

- **Stage:** Early-stage customer (typically < 12 months).
- **Plan Setup:** Month-to-Month billing with Paperless Billing enabled.
- **Payment Method:** Pays manually via Electronic Check.
- **Value Gap:** Subscribed to expensive core internet services (like Fiber Optic) but lacks supplementary retention services (e.g., Device Protection, Tech Support).



4. Revenue Impact & Actionable Recommendations

Customer churn directly impacts MRR. By utilizing our 79.91% accurate Prediction System, TEYZIX CORE can proactively identify High-Risk customers and apply the following strategies:

- **Targeted Interventions (Contract Upgrade):** Offer a 10-15% discount or a free month of 'Tech Support' to High-Risk customers who agree to switch from a Month-to-Month contract to a 1-Year contract.
- **Reduce Payment Friction:** Incentivize customers to set up automatic payments (Credit Card/Bank Transfer) to reduce the mental friction of manual monthly payments via electronic checks.
- **Segment Protection:** Engage 'High Value' segment customers who fall into the 'Medium Risk' category with premium loyalty programs. Protecting this specific segment ensures the most significant portion of the revenue stream remains secure.

Customer Segmentation Distribution

